

Sliding Mode Control for Electromagnetic Levitation System Based on Feedback Linearization

Muhammad Aamir Akram
Department of Electronic Engineering
The Islamia University of Bahawalpur
Bahawalpur, Pakistan
amir.akram@iub.edu.pk

Ijlal Haider, Habib-Ur-Rehman Khalid
MEDICS Lab., Department of Electrical Engineering
The University of Lahore
Lahore, Pakistan
ijlal.haider@ee.uol.edu.pk

Vali Uddin
Hamdard University
Karachi, Pakistan
Vali.Uddin@hamdard.edu.pk

Abstract - Magnetic levitation system has developed into a significant consideration in designing systems which require low losses due to friction and low energy consumption. In this paper the levitation inflexibility for suspending a steel ball and its distinctive application to wipe out friction and build machining accuracy is talked about. The procedures of feedback linearization and sliding mode control are proposed in this paper to accomplish relentless levitation. The procedure of feedback linearization is utilized as a part of this paper to change a nonlinear electromagnetic levitation framework into a linear framework. The proposed controller of sliding mode control is received for enhancing the power and robustness of the electromagnetic levitation system in the vicinity of commotions. Simulation results of sliding mode control demonstrate that the system has more prevailing and powerful bent of stifling disturbance and high inflexibility as equated with that utilizing traditional PID control strategy.

Keywords—Control Systems, Sliding Mode Control, Feedback Linearization, Magnetic Levitation

I. INTRODUCTION

Levitation is defined as the stable equilibrium of a body without contact with the earth. The utilization of magnetic power to suspend objects with no obvious method for backing is termed as magnetic levitation or MAGLEV. In view of the rule of operation, magnetic levitation might comprehensively be arranged into two sorts, electrodynamic levitation and electromagnetic levitation [1-4]. This section summarizes the introduction, advantages, applications, fundamental issues, linearization technique and existing SMC controller which is used for MAGLEV system.

With the improvement of control technology, magnetic levitation issues have pulled in numerous scientists and engineers consideration inferable from its friction free dynamic movements. To represent numerous fundamental standards of electrical and electronic engineering a MAGLEV metallic ball framework has been a subject of significant enthusiasm, for example, electromagnetic, circuit design, and control calculations. The MAGLEV framework is nonlinear

type and in open loop it is insecure. The system functioning would be vigorously influenced by different disturbances. These testing and intriguing attributes accelerate the design of different controllers keeping in mind the end goal to enhance the system efficiency and performance.

In Magnetic Levitation systems, closed loop consists of actuators, sensors, electromagnet and controllers. Actuator is used as voltage to current convertor which changes over the output voltage from controller into input current for the electromagnet. The explanation behind utilizing this system is to isolated power amplifier from controller. Sensor is used to sense the position of metallic ball which is based on Hall Effect. Electromagnet and a steel ball is our MagLev plant and controller design is our main goal for this paper to enhance the performance and efficiency of MagLev framework.

The objective of the paper is to analyse the MagLev system mathematically and to use the technique of feedback linearization to linearize the system and then design a sliding mode controller to enhance the functioning and efficiency of the framework in the presence of disturbances and uncertainties. So, in the overall procedure of controller design the technique of sliding mode control is used after linearizing the framework by using the technique of feedback linearization.

II. LITERATURE SURVEY AND BACKGROUND

The research involved in this paper centrally focused on modelling of system, linearization techniques and designing of controller. In this section the literature review of the work done in the related field is discussed.

Ahmad El Hajjaji and M Ouladsine [1] analysed modelling and nonlinear control of MAGLEV framework. In this paper author proposed a nonlinear model for magnetic levitation system and utilizing this model designed a nonlinear control law in view of differential geometry.

Ximin Shan, Shih-Kang Kuo, Jihua Zhang and Chia Hsiang Manq [5] proposed ultra-precision movement control

of a various degrees of freedom magnetic suspension stage. The control architecture designed in this paper is robust nonlinear controller and consists of three parts which are: 1) feedback linearization, 2) force distribution, and 3) H_∞ robust controller for each degree of freedom (DOF) of movement.

Regis Campos Fama, Renato Vilela Lopes and Anderson de Paulo Milhan [6] considered predictive control of a magnetic levitation framework with unequivocal treatment of operational limitations. In this paper the optimal control arrangement is actualized in the receding-horizon strategy, in which the optimization is rehashed at each taking so as to sample moment, by taking into record the new sensor reading.

Subrata Banerjee, Jayanta Pal and Dinkar Parsad [7] considered the execution and efficiency investigation of the controller of an attraction type levitation framework under parametric change. In this work author proposed a cascaded lead compensated controller which consist of an inner current loop and outer position loop and carried out this controller to improve stability and efficiency of single magnet based single axis levitation system.

Shafayet Hossain [8] proposed a pattern of a robust controller for a magnetic levitation system. To overcome nonlinearity problems and achieve stability, author design a robust controller using H-infinity optimization.

Sliding Mode Control (SMC) hypothesis was established and advanced in the previous Soviet Union as a variable structure control framework. SMC is fundamentally an outcome of discontinuous control. SMC was discovered at the beginning of sixties when engineers were looking for ways to design robust control laws. SMC theory initially showed up outside Russia in the mid-1970s when a book by Itkis in 1976 and a review paper by Utkin [13] in 1977 were distributed in English. The SMC reachability condition depends on the hypothesis of Russian mathematician "Lyapunov" on stability of nonlinear systems [9].

III. SYSTEM DESCRIPTION AND CONTROLLER DESIGN

Magnetic Levitation Systems are finding application in several areas. Fig. 1 shows physical model of MAGLEV system. In order to study MAGLEV System a process is considered which consist of a magnetic ball suspension framework. The object of which is to keep a metal ball suspended in air by adjusting the field strength of an electromagnet. This section describes the basics of system, system dynamics, its modelling and essentials of magnetic levitation system.

Following equations shows the dynamics of electromagnet by using Newton's law of motion:

$$m\ddot{z} = -F(i, z) + mg \quad (1)$$

$$m\ddot{z} = -k \left[\frac{i(t)}{z(t)} \right]^2 + mg \quad (2)$$

Where

$F(i, z)$ is electromagnet force;

m is the mass of the levitated ball;

g denotes acceleration due to gravity;

z is the distance of the ball from electromagnet;

I is the current through the electromagnet;

K is a coefficient;

μ_0 is permeability of free space;

A is the field area of electromagnet core.

Equation (2) describes the complete model of electromagnetic levitation system and indicates that the system is nonlinear due to the $\left[\frac{i(t)}{z(t)} \right]^2$ term. We have used the technique of feedback linearization to linearize the electro MAGLEV system which is discussed next.

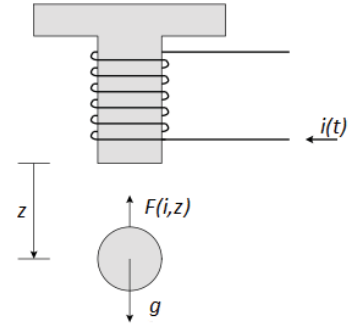


Fig. 1. Physical Model of MagLev System

Feedback linearization is a typical methodology utilized as a part of controlling nonlinear systems. The linear system is acquired through a change of variables and a suitable control input. Feedback linearization [10] may be applied to the nonlinear systems of the form:

$$\dot{x} = f(x) + g(x)u \quad (3)$$

$$y = h(x) \quad (4)$$

The main purpose is to develop a feedback control input

$$u = \alpha(x) + \beta(x)v \quad (5)$$

To implement feedback linearization the first step is to find a state-space transformation and to find a state feedback control law. From (2) the states of this system are defined as:

$$x_1 = z \quad (6)$$

$$x_2 = \dot{z} \quad (7)$$

$$x = [x_1 \ x_2]^T = [z \ \dot{z}]^T \quad (8)$$

Where x_1 and x_2 are states of the system which are equal to the air gap and rate of change of air gap with respect to time respectively and x in (8) is chosen as the state variable, then the representation of dynamic system in nonlinear state-space equation as:

$$\dot{x}_1 = \dot{z} = x_2 \quad (9a)$$

$$\dot{x}_2 = \dot{z} = g - \frac{k}{m} \left[\frac{t}{x_1} \right]^2 \quad (9b)$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} x_2 \\ -\frac{k}{m} \left(\frac{t}{x_1} \right)^2 + g \end{bmatrix} \quad (10)$$

$$y = x_1$$

According to

$$\dot{z} = \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = [f(x) + g(x)\alpha(x) + g(x)\beta(x)v] \quad (11a)$$

and by placing the values of $\alpha(x)$ and $\beta(x)$ from Lie Derivative in above equation,

$$\dot{z} = \begin{bmatrix} x_2 \\ g \end{bmatrix} + \begin{bmatrix} 0 \\ -\frac{k}{m} \left(\frac{1}{x_1^2} \right) \end{bmatrix} \left[\frac{mgx_1^2}{k} + \left(-\frac{mx_1^2}{k} \right) \right] \quad (11b)$$

By solving and simplifying the above equation, constants and higher order terms are cancelled out and the obtained linear state space equation is as,

$$\dot{z} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} z + \begin{bmatrix} 0 \\ 1 \end{bmatrix} v \quad (12a)$$

$$y = [1 \ 0]z \quad (12b)$$

The system (12a, 12b) is linear and controllable and it can be stabilized by using different types of controllers. For the efficient close loop performance of the system, there is a need of effective and robust controller for this system.

The controller design is subdivided into two steps [9, 11]. These steps include the designing of sliding surface and designing of a control law. Fig. 2 describes the schematic of MagLev system with SMC controller where $\tilde{z}$ is the desired input signal and z is the actual output of the framework.

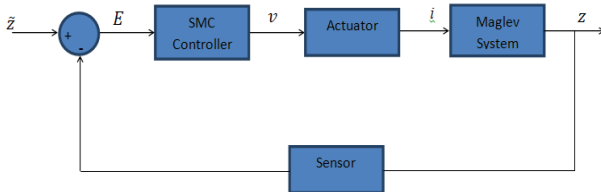


Fig.2. Schematic of MagLev System with SMC

The sliding surface for a second order time-invariant framework is a switching line in phase space. It is designed in the form of a line or hyperplane in state-space to compel a sliding motion.

The linearized model is given in (12a) for electromagnetic levitation system. Let the desired output be

$$\tilde{z} = [\tilde{z}_1 \ \tilde{z}_2]^T \quad (13)$$

Where $\tilde{z}_1$ and $\tilde{z}_2$ are desired levitation height and rate of change of levitation height with respect to time respectively.

The actual output of the electromagnetic levitation system (EMLS) is

$$z = [z_1 \ z_2]^T \quad (14)$$

The levitation position of the suspended ball is defined as

$$E = \tilde{z} - z = [e \ \dot{e}]^T \quad (15)$$

The sliding manifold or surface is formulated as

$$S = CE = C[e \ \dot{e}]^T \quad (16)$$

Where S is sliding surface and $C = [c_1 \ c_2]$ is the coefficient of sliding surface which is selected so as the rate of change of levitation height with respect to time is automatically adjustable as the disturbances varies and c_1, c_2 are selected so as to maintain desired levitation height according to the requirement and to bring the system towards the sliding surface in a finite time.

$$S = [c_1 \ c_2][e \ \dot{e}]^T \quad (17)$$

So,

$$S = c_1 e + c_2 \dot{e} \quad (18)$$

In SMC, the motion is independent of the control. So the control must be designed in such a way that it derives the trajectories towards the sliding surface and maintain these trajectories on the sliding surface once it has been achieved. After the motion of the system enters into the sliding mode, it must satisfy $S=0$, so the derivative of ' S ' is also equal to zero, that is

$$\dot{S} = C\dot{E} = C(\dot{z} - \dot{z}) \quad (19)$$

From (12a)

$$\dot{z} = Az - Bv_{eq} \quad (20)$$

By placing the value of $\dot{z}$ in (19) and solving it for v_{eq}

$$\dot{S} = C\dot{z} - CAz - CBv_{eq} \quad (21)$$

$$v_{eq} = (CB)^{-1}(C\dot{z} - CAz) \quad (22)$$

v_{eq} is the equivalent control which is used to provide exact sliding surface and brings the system's movement towards the sliding surface.

Reachability condition necessitates that system movement must arrive in finite time at sliding surface. This condition is necessary to be sure that sliding mode starts at some time $t > 0$, irrespective of the initial state $x(0)$, it is necessary that the state trajectory is always moving towards $S = 0$, whenever S is not zero, that is it satisfies $S\dot{S} \leq 0$ which called reaching condition. Lyapunov method is used to verify

reachability condition [12]. Lyapunov function is stated as follows:

$$v = \frac{1}{2}s^2 \quad (23)$$

Which is positive semi-definite and its derivative to satisfy reachability condition is required as

$$\dot{v} = s\dot{s} \leq 0 \quad (24)$$

A controller must be designed in such a way to satisfy the above reachability condition

$\dot{v} \leq 0$ for all values of time. Then the response of the controller must be guaranteed to reach the sliding surface.

By selecting the exponential reaching law as

$$\dot{s} = -\varepsilon \text{sgn}(s) - ks \quad (25)$$

Where the constants are ε , k and the values of ε , $k > 0$. The values of ε , k are very significant for the behaviour and properties of the system. So the selection of ε , k should be proper to accelerate the movement and avoid chattering [1]. By solving and substituting equations the control law is obtained as

$$v = (CB)^{-1}(C\dot{d} - CAx) + (CB)^{-1}(-\varepsilon \text{sgn}(s) - ks) \quad (26)$$

Equation (26) is the addition of two parts,

$$v_{eq} = (CB)^{-1}(C\dot{d} - CAx) \quad (27)$$

and

$$v_d = (CB)^{-1}(-\varepsilon \text{sgn}(s) - ks) \quad (28)$$

The general form of control law is written as

$$v = v_{eq} + v_d \quad (29)$$

Where v_d is the discontinuous part of the control law which ensures the reachability condition, i.e., convergence of system motion to the sliding surface is in a finite time. Combination of equivalent control with switching control should be applied to obtain strong robustness due to the presence of uncertainties and external interferences. The parameters ε , k , C_1 and C_2 can be properly selected to maintain position error and to achieve desired response. The approximate values of these parameters can be obtained by hit and trial method.

In the vicinity of diverse flaws in switching process, for example, time delays in switching process and little time constants in the actuators, the feedback control law creates a specific dynamic behaviour because of discontinuity in the surface's region, which is normally referred to as chattering. The most regularly referred way to decrease the impacts of chattering has been the alleged piecewise linear or smooth guess of switching component in a boundary layer in the vicinity of sliding surface [3, 4]. The subsequent control with chattering suppression is

$$v = (CB)^{-1}(C\dot{d} - CAx) + (CB)^{-1}(-\varepsilon \text{sat}(s) - ks) \quad (30)$$

IV. SIMULATION AND RESULTS

In this segment, we exhibit some test results demonstrating the object's behaviour in levitation utilizing a proportional-integral-derivative (PID) controller and sliding mode controller. MATLAB has been adopted in analysis and design of control system for its simplicity and comprehensiveness by researchers. Step by step results are taken by running Simulink blocks.

We have observed the response of both controllers in the presence of disturbance and it is clear that the performance of PID controller is deteriorated as compared to SMC controller. It also shows that linear controller gives good results in safe and protected environments. SMC controller shows much more better and robust performance than conventional PID controller.

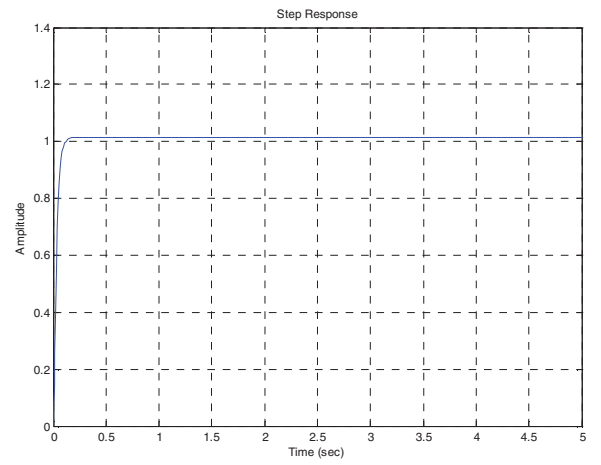


Fig. 3. Step Response of SMC Controller

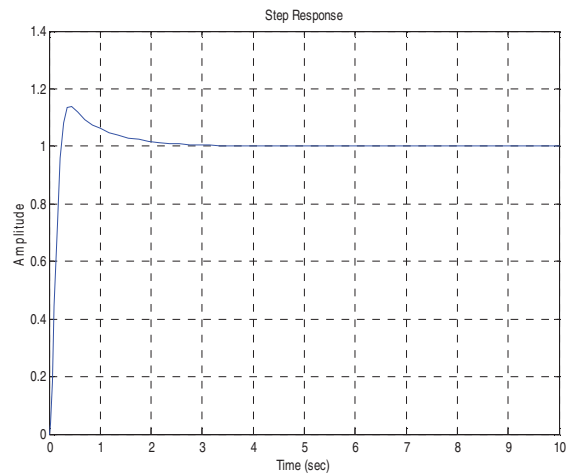


Fig. 4. Step Response of PID Controller

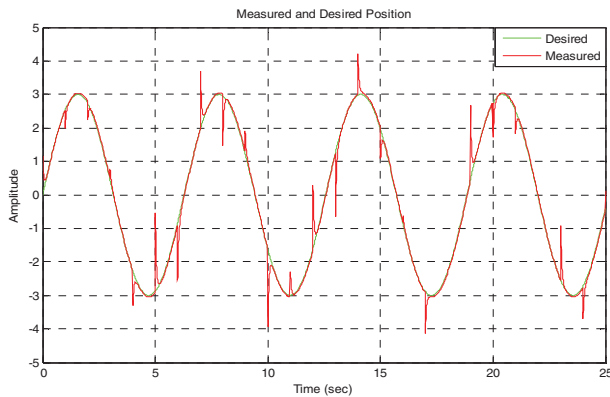


Fig. 5. Response of SMC with Disturbance

V. CONCLUSION

In this paper the primary accentuation is given on the technique of feedback linearization to linearize the electro MAGLEV system and sliding mode controller's designing to see the functioning of designed control system in the presence of external disturbances. The functioning of proposed control technique (SMC) is compared with the conventional PID control method. So in the presence of disturbances it is useful to see the functioning of conventional PID controller. The comparison of sliding mode control with PID controller realizes that the system with SMC controller shows not only the rapid response but also disturbance rejection and interference suppression. Parameter tuning techniques can be used to improve the performance of SMC controller and to remove the steady state error of step response as shown in Fig.3.

REFERENCES

- [1] Ahmad El Hajjaji and M Ouladsine, "Modelling and nonlinear control of magnetic levitation system," IEEE Transaction on Industrial Electronics, Vol.48, No.4, 2001.
- [2] Walliam G. Hurley and Werner H. Woffle, "Electromagnetic design of a magnetic suspension system," IEEE Transaction on Education, Vol. 40, No. 2, 1997.
- [3] Faa-jeng Lin, Li-Tao Teng and Po-Huang Shieh, "Intelligent sliding mode control using RBFN for magnetic levitation system," IEEE Transaction on Industrial Electronics, Vol. 54, No. 3, 2007.
- [4] Sung Jun Joo and Jin H. Seo, "Design and analysis of the nonlinear feedback linearizing control for an electromagnetic suspension system," IEEE Transaction on Control System Technology. Vol. 5, No.1, 1997.
- [5] Ximin Shan, Shih-Kang Kuo, Jihua Zhang and Chia-Hsiang Menq, "Ultra Precision Motion Control of a Multiple Degrees of Freedom Magnetic Suspension Stage," IEEE/ASME Transactions on Mechatronics, Vol. 7, No. 1, 2002.
- [6] Régis Campos Fama, Renato Vilela Lopes, Anderson de Paulo Milhan and Roberto Kawakami Harrop Galvão, "Predictive Control of a Magnetic Levitation System With Explicit Treatment of Operational Constraints," 18th International Congress of Mechanical Engineering, November 6-11, 2005.
- [7] Subrata Banerjee, Jayanta Pal, Dinkar Parsad, "Performance study of the controller of an attraction type levitation system under parametric change," Journal of Electrical Systems Vol. 6-3, Pg. 377-394, 2010.
- [8] Shafayet Hossain, Design of a Robust Controller for a Magnetic Levitation System. Wichita State University, Wichita, KS.
- [9] Wilfrid Perruquetti, Jean Pierre Barbot, "Sliding Mode Control in Engineering," Marcel Dekker, Inc New York 2002.
- [10] Liu DE-sheng, Li Jie, Zhang Kun, "The Design of the Nonlinear Suspension Controller for EMS MagLev Train Based on Feedback Linearization," Journal of National University of De Fense Technology, Vol.27, No.2, Pg. 96-101, 2005.
- [11] K. D. Young, V. I. Utkin, and U. Ozguner, "A control engineer's guide to sliding mode control," IEEE Transactions on Control Systems Technology, Vol. 7 No. 3, Pg. 328-342, 1999.
- [12] Zhang T P, "Adaptive fuzzy sliding mode control based on a modified Lyapunov function," Acta Automatica Sinica, Vol.1, No.28, Pg. 137-142, 2002.
- [13] V. I. Utkin, "Variable structure systems with sliding modes," IEEE Transaction on Automatic Control, Vol. 22, No. 2, Pg. 212-222, 1977.
- [14] J. J. Slotine, Applied Nonlinear Control. Englewood Cliffs, NJ: Prentice Hall, 1991.
- [15] Hassan K. Khalil, Nonlinear Systems. NJ 07458: Prentice-Hall, November 2008.